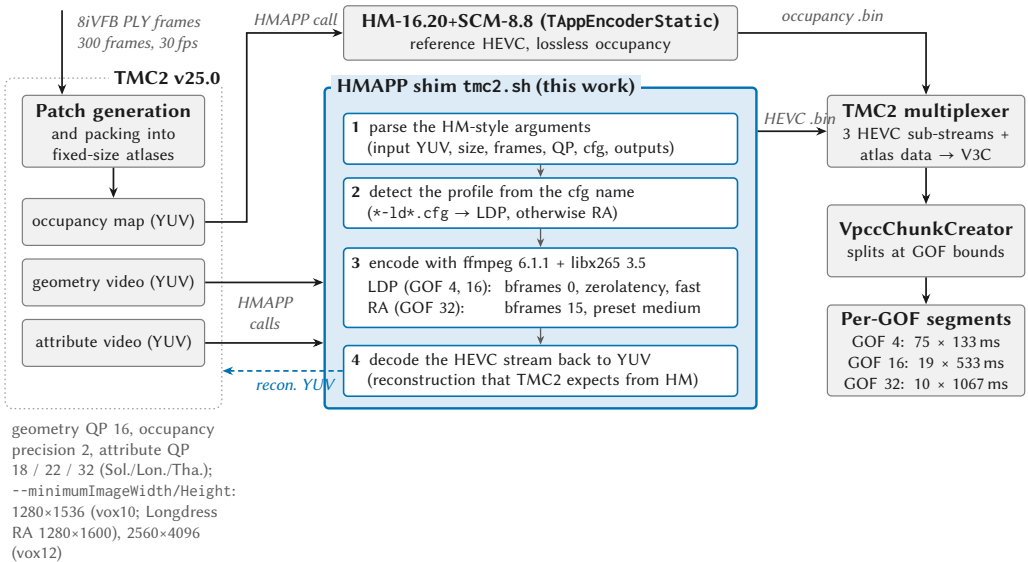
8iVFB PLY frames
300 frames, 30 fps
TMC2 v25.0
Patch generation
and packing into
fixed-size atlases
occupancy map (YUV)
geometry video (YUV)
attribute video (YUV)
HMAPP call
HMAPP
calls
recon. YUV
HM-16.20+SCM-8.8 (TAppEncoderStatic)
reference HEVC, lossless occupancy
occupancy .bin
HMAPP shim tmc2.sh (this work)
1 parse the HM-style arguments
(input YUV, size, frames, QP, cfg, outputs)
2 detect the profile from the cfg name
(*-ld*.cfg → LDP, otherwise RA)
3 encode with ffmpeg 6.1.1 + libx265 3.5
LDP (GOF 4, 16): bframes 0, zerolatency, fast
RA (GOF 32): bframes 15, preset medium
4 decode the HEVC stream back to YUV
(reconstruction that TMC2 expects from HM)
HEVC .bin
TMC2 multiplexer
3 HEVC sub-streams +
atlas data → V3C
VpccChunkCreator
splits at GOF bounds
Per-GOF segments
GOF 4: 75 × 133 ms
GOF 16: 19 × 533 ms
GOF 32: 10 × 1067 ms
geometry QP 16, occupancy
precision 2, attribute QP
18 / 22 / 32 (Sol./Lon./Tha.);
--minimumImageWidth/Height:
1280×1536 (vox10; Longdress
RA 1280×1600), 2560×4096
(vox12)
new (this work)
upstream, patched by us
upstream / third party, unmodified